

CS 310 : Automata Theory

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- ▶ For $F = \{a^i b^j c^k \mid i = 1 \Rightarrow j = k\}$, the **exactly one** a which forces $j = k$ is at the front, used by the pump window
- ▶ the pumping lemma only needs **one** safe split to exist
- ▶ we can pick the **pump a** option, which is the safe $i \neq 1$ case,
- ▶ the unsafe option (pumping the b 's) is never forced

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- ▶ In our earlier version of pumping lemma, $u, v = \epsilon$.
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 - ▶ Is this an iff condition?

Apply to $F = \{a^i b^j c^k \mid i=1 \Rightarrow j=k\}$

The subword pumping lemma as stated: for each p , find u, w, v with $uwv \in F$, $|w| \geq p$, such that every decomposition $w = xyz$ fails to pump.

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This holds for every p , so by the subword pumping lemma, F is not regular.

A necessary and sufficient Pumping Lemma

(Jaffe, 1978)

Jaffe's Pumping Lemma

Theorem

$L \subseteq \Sigma^*$ is regular **if and only if** there exists $n \geq 1$ such that: for every word $y \in \Sigma^n$, there exist words u, v, w factorizing y as $y = uvw$ with $v \neq \varepsilon$, and for **every** $z \in \Sigma^*$ and every $i \geq 0$,

$$uv^i wz \in L \iff uvwz \in L.$$

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- ▶ L regular $\Rightarrow$ pumping property holds
- ▶ pumping property holds $\Rightarrow L$ regular: build a DFA directly from the property.

Necessity: L regular $\Rightarrow$ property holds

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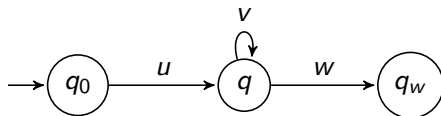
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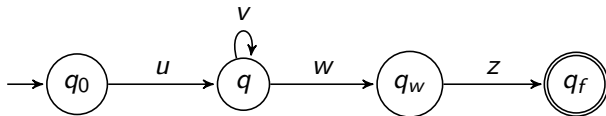
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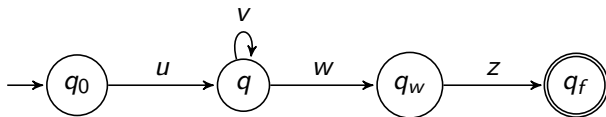
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The loop on q means u and uv, uv^i all reach the *same* state. $uwz, uvwz, uv^2wz$ are all either accepted or not for any continuation z .

Sufficiency: property holds $\Rightarrow L$ regular

Let L be a language.

Assume $\exists n \geq 1$ such that for all words $y \in \Sigma^n$ there exist words u, v, w factorizing $y = uvw$ with $v \neq \epsilon$ and for all words $z \in \Sigma^*$,

$$uvwz \in L \Leftrightarrow uv^i wz \in L$$

for all $i \geq 0$. Then L is regular.

Construct a DFA M for L .

An Example

$L = \{w \in \{a, b\}^* : w \text{ contains } ab \text{ as a subword}\}$. We know L is regular.

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- ▶ Consider $n = 2$. $\Sigma^2 = \{aa, ab, ba, bb\}$.
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 - ▶ $v = b$ ($u = a, w = \varepsilon$): $uwz = a \notin L$.
 - ▶ $v = ab$ ($u = \varepsilon, w = \varepsilon$): $uwz = \varepsilon \notin L$.
 - ▶ So $n \neq 2$.

An Example

$L = \{w \in \{a, b\}^* : w \text{ contains } ab \text{ as a subword}\}$

- ▶ Consider $n = 3$. $\Sigma^3 = \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$.
- ▶ For each $y \in \Sigma^3$, we can check if the property holds.

Checking all 8 words of length 3

y	u	v	w	uw	uvw	$uv^i w \ (i > 1)$
aaa						
aab						
aba						
abb						
baa						
bab						
bba						
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bab	ϵ	b	ab	ab	bab	$b^i ab$
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- ▶ Note that $|uw| \leq 2$
- ▶ We can start building the automaton by first generating words of length < 3 , and use the above property to generate words of length 3

Building the automaton

- ▶ Words of length 3 contain a pump : they can be factorized as uvw such that for all $z \in \Sigma^*$, $uwz \in L$ iff $uv^i wz \in L$ for all $i \geq 1$
- ▶ So acceptance of uvw is same as that of uw when $z = \epsilon$
- ▶ Note that $|uw| \leq 2$
- ▶ We can start building the automaton by first generating words of length < 3 , and use the above property to generate words of length 3
- ▶ Then use the same property to generate words of length > 3

Building the automaton

- ▶ Words of length < 3 : ϵ , a , b , aa , bb , ba , ab .
- ▶ Consider words of length 3

Building the automaton

- ▶ Words of length < 3 : ϵ , a , b , aa , bb , ba , ab .
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 - ▶ aaa is accepted iff aa is (check the table)
 - ▶ aab is accepted iff ab is, and so on

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So $x = aaaa$ behaves exactly like aa : $xb \in L \iff aab \in L$.

To continue on a brighter day!